Ex. 1.

Q.2

Year	t	X_t Crude oil	Price of Y_t leaded regular
1995		10.38	0.57
1996		10.89	0.59
1997		11.96	0.62
1998	1	19.46	0.63
1999	2	17.72	0.86
2000	3	28.07	1.19
2001	4	35.24	1.31
2002	5	31.87	1.22
2003	6	28.99	1.16
2004	7	28.63	1.13
2005	8	26.75	1.12
2006	9	14.55	0.86
2007	10	17.90	0.90
2008	11	14.71	0.90

(a), (b) and (c)

Year	Y_t Price	$\hat{Y}_t$ Naive	$\hat{Y}_t$ MA(3)	$\hat{Y}_t$ Trend	$\hat{Y}_t$ Causal
97	0.62				
1998	0.63	0.62	0.59	1.023	0.931
1999	0.86	0.63	0.61	1.024	0.887
2000	1.19	0.86	0.70	1.025	1.146
2001	1.31	1.19	0.89	1.026	1.325
2002	1.22	1.31	1.12	1.027	1.241
2003	1.16	1.22	1.24	1.028	1.169
2004	1.13	1.16	1.23	1.029	1.16
2005	1.12	1.13	1.17	1.030	1.113
2006	0.86	1.12	1.14	1.031	0.808
2007	0.90	0.86	1.04	1.032	0.892
2008	0.90	0.90	0.96	1.033	0.812

Please see calculation on page (2).

MA(3) $\hat{Y}_t$ calculation

$\frac{0.57+0.59+0.62}{3} = 0.593 \simeq 0.59$

$\frac{0.59+0.62+0.63}{3} = 0.606 \simeq 0.61$

$\frac{0.62+0.63+0.86}{3} = 0.703 \simeq 0.70$

$\frac{0.63+0.86+1.19}{3} = 0.893 \simeq 0.89$

$\frac{0.86+1.19+1.31}{3} = 1.12$

$\frac{1.19+1.31+1.22}{3} = 1.24$

$\frac{1.31+1.22+1.16}{3} = 1.23$

$\frac{1.22+1.16+1.13}{3} = 1.17$

$\frac{1.16+1.13+1.12}{3} = 1.136 \simeq 1.14$

$\frac{1.13+1.12+0.86}{3} = 1.036 \simeq 1.04$

$\frac{1.12+0.86+0.90}{3} = 0.96$

Trend $\hat{Y}_t = 1.022 + 0.001 \times t$, $t = 1, 2, \ldots, 11$ (Year 98, 99, ..., 08)

$\hat{Y}_{98} = 1.022 + 0.001(1) = 1.023$

$\hat{Y}_{99} = 1.022 + 0.001(2) = 1.024$

$\hat{Y}_{00} = 1.022 + 0.001(3) = 1.025$

$\hat{Y}_{01} = 1.022 + 0.001(4) = 1.026$

$\hat{Y}_{02} = 1.022 + 0.001(5) = 1.027$

$\hat{Y}_{03} = 1.022 + 0.001(6) = 1.028$

$\hat{Y}_{04} = 1.022 + 0.001(7) = 1.029$

$\hat{Y}_{05} = 1.022 + 0.001(8) = 1.030$

$\hat{Y}_{06} = 1.022 + 0.001(9) = 1.031$

$\hat{Y}_{07} = 1.022 + 0.001(10) = 1.032$

$\hat{Y}_{08} = 1.022 + 0.001(11) = 1.033$

Causal $\hat{Y}_t = 0.444 + 0.025 X_t$ (Year 98, 99, ..., 08)

$\hat{Y}_{98} = 0.444 + 0.025(19.46) = 0.9305 \simeq 0.931$

$\hat{Y}_{99} = 0.444 + 0.025(17.72) = 0.887$

$\hat{Y}_{00} = 0.444 + 0.025(28.07) = 1.1457 \simeq 1.146$

$\hat{Y}_{01} = 0.444 + 0.025(35.24) = 1.325$

$\hat{Y}_{02} = 0.444 + 0.025(31.87) = 1.24075 \simeq 1.241$

$\hat{Y}_{03} = 0.444 + 0.025(28.99) = 1.16875 \simeq 1.169$

$\hat{Y}_{04} = 0.444 + 0.025(28.63) = 1.15975 \simeq 1.16$

$\hat{Y}_{05} = 0.444 + 0.025(26.75) = 1.11275 \simeq 1.113$

$\hat{Y}_{06} = 0.444 + 0.025(14.55) = 0.80775 \simeq 0.808$

$\hat{Y}_{07} = 0.444 + 0.025(17.90) = 0.8915 \simeq 0.892$

$\hat{Y}_{08} = 0.444 + 0.025(14.71) = 0.81175 \simeq 0.812$

1.(a) $e_t = Y_t - \hat{Y}_t$

Year	Y_t	Naive e_t	MA(3) e_t	Trend e_t	Causal e_t
1998	0.63	0.01	0.04	-0.393	-0.301
1999	0.86	0.23	0.25	-0.164	-0.027
2000	1.19	0.33	0.49	0.165	0.044
2001	1.31	0.12	0.42	0.284	-0.015
2002	1.22	-0.09	0.10	0.193	-0.021
2003	1.16	-0.06	-0.08	0.132	-0.009
2004	1.13	-0.03	-0.10	0.101	-0.03
2005	1.12	-0.01	-0.05	0.090	0.007
2006	0.86	-0.26	-0.28	-0.171	0.052
2007	0.90	0.04	-0.14	-0.132	0.008
2008	0.90	0.00	-0.06	-0.133	0.088

	Naive	MA(3)	Trend	Causal
MSE	0.0234	0.055	0.039	0.0096
RMSE	0.1532	0.234	0.197	0.098
MAE	0.1073	0.181	0.178	0.055
MAPE	0.10545 $\simeq$ 0.1055 or 10.55%	17.24%	19.14%	7.11%

↑

Causal model is the best from all criteria.

(e)

Year	Naive $\hat{Y}_t$	MA(3) $\hat{Y}_t$	Trend $\hat{Y}_t$
2009	0.90	$0.886 \simeq 0.89$	1.034
2010	0.90	$0.896 \simeq 0.90$	1.035

Q.3 $DPS_t = Y_t$ $EPS_t = X_t$

(a)

Year	Y_t	Naive $\hat{Y}_t$	e_t	MA(3) $\hat{Y}_t$	e_t
97	0.53				
1998	0.57	0.53	0.04	0.53	0.04
1999	0.72	0.57	0.15	0.54	0.18
2000	0.87	0.72	0.15	0.61	0.26
2001	0.87	0.87	0.00	0.72	0.15
2002	0.87	0.87	0.00	0.82	0.05
2003	0.87	0.87	0.00	0.87	0.00
2004	0.87	0.87	0.00	0.87	0.00
2005	0.87	0.87	0.00	0.87	0.00
2006	0.87	0.87	0.00	0.87	0.00
2007	0.90	0.87	0.03	0.87	0.03

(b) & (c) Trend : $\hat{Y}_t = 0.694 + 0.024 \times t$

Causal : $\hat{Y}_t = 0.961 - 0.091 X_t$

Year	Y_t	Trend $\hat{Y}_t$	e_t	Causal $\hat{Y}_t$	e_t
1998	0.57	0.718	-0.148	0768	-0.198
1999	0.72	0.742	-0.022	0.717	0.003
2000	0.87	0.766	0.104	0.811	0.059
2001	0.87	0.790	0.08	0.860	0.010
2002	0.87	0.814	0.056	0.906	-0.036
2003	0.87	0.838	0.032	0.878	-0.008
2004	0.87	0.862	0.008	0.848	0.022
2005	0.87	0.886	-0.016	0.881	-0.011
2006	0.87	0.910	-0.040	0.845	0.025
2007	0.90	0.934	-0.034	0.768	0.132

Q.3 (d)

	Naive	MA(3)	Trend	Causal
MAE	0.037	0.071	0.054	0.0504
MAPE	0.0484 or 4.84%	0.0710 or 7.10%	0.0714 or 7.14%	0.0695 or 6.948%
MSE	0.00475	0.01275	0.04684	0.0063
RMSE	0.0689	0.1129	0.2164	0.0793

↑

Naive model is the best by all criteria.

(e)

Year	Y_t	Naive	e_t	MA(3)	e_t	Trend	e_t	Causal	e_t
07	0.9								
2008	1.15	0.9	0.25	0.88	0.27	0.958	0.192	0.717	0.433
2009	1.20	1.15	0.05	0.97	0.23	0.982	0.218	0.703	0.497
2010	1.20	1.20	0	1.08	0.12	1.006	0.194	0.791	0.409

(f)

	Naive	MA(3)	Trend	Causal
MAE	0.1	0.183	0.2013	0.4463
MAPE	0.0864 or 8.635%	0.1552 or 15.519%	0.1701 or 17.01%	0.37717 or 37.717%
MSE	0.0216	0.0467	0.04067	0.2006
RMSE	0.1472	0.2162	0.2017	0.4479

↑

Again, Naive model is the best.

(g) Naive model is the best by all criteria. The reason is due to the constancy of the dividend payment over the period.